

COMSATS University Islamabad, Vehari Campus

Information Communication Technology (CSC101)

Lab 08

Topic: Problem Solving with Repetition Structure

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Objective

- Understand repetition (looping) structures
 - Apply loops to solve real-world problems
 - Use nested loops and decision structures
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1. What is a Loop?

A loop is used to repeat a set of instructions until a condition becomes true.

✓ Why use loops?

- Avoid code repetition
 - Handle unknown number of inputs
 - Perform repetitive calculations efficiently
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2. Types of Loops

Counter-Controlled Loop

- Runs a fixed number of times
- Uses a counter variable

Sentinel-Controlled Loop

- Runs until a condition is met
 - Number of repetitions is unknown
-

3. Loop Control (I.T.E.M Rule)

Step	Description
Initialize	Set starting value
Test	Check condition
Execute	Run statements
Modify	Update variable

Note: Missing any step may cause infinite loop

4. Infinite Loops

A loop that never ends due to: - Missing update - Wrong condition - Incorrect initialization

5. Nested Structures

- Loop inside loop
 - Decision inside loop
 - Loop inside decision
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Solved Lab Activities

Activity 1 – Loop Control & Validation

Concept

- Input validation using loop
- Counter-controlled loop

Explanation

- The loop continues until the input becomes correct
- A loop control variable (Count) is used

Key Idea

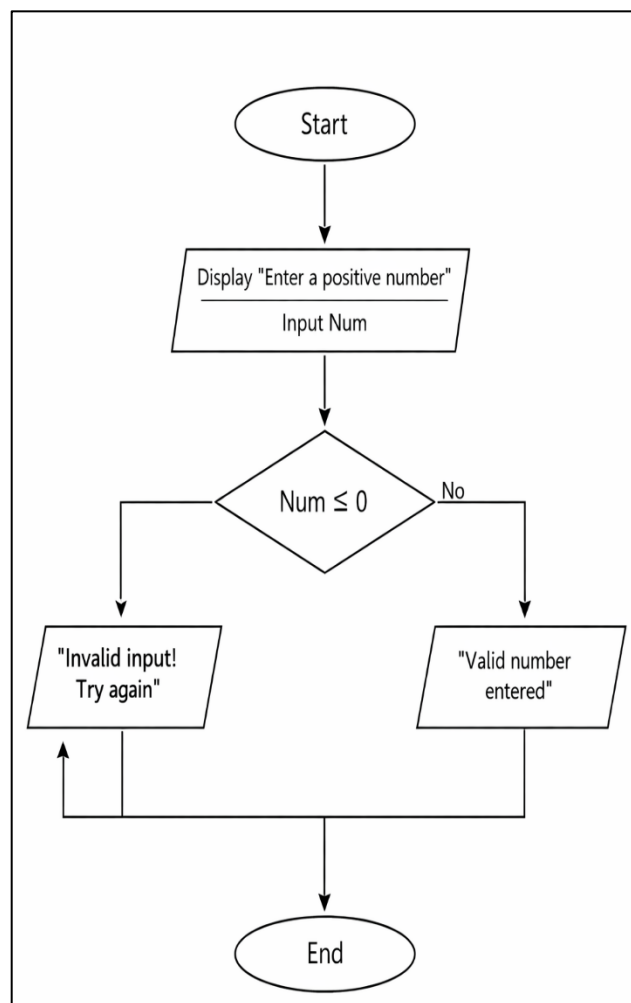
A loop requires three things:

- Initialize
- Test
- Modify

Infinite Loop Cases

- The variable is not updated
- The condition never becomes false

Flowchart:



Activity 2 – Sentinel Controlled Loop

Concept

- The user enters a "special value" (sentinel value)
- The loop continues until the sentinel value is entered

Example

- The user enters numbers
- When the user enters 0 → the loop stops

Pseudocode

Input Data

While Data != 0

 Process Data

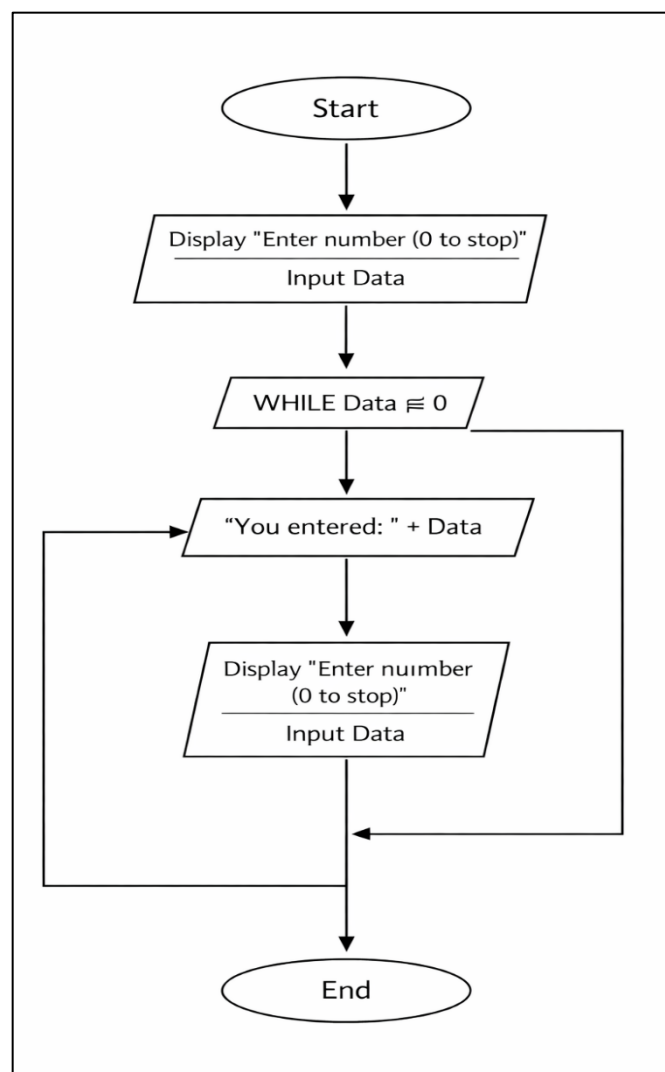
 Input Data

End While

Key Idea

- It is used to handle an unknown number of inputs.

Flowchart:



Activity 3 – Counting Events

Concept

- Counting is done by checking a condition inside the loop

Example

- Counting positive numbers

Pseudocode

Count = 0

Input Data

While Data != 0

 If Data > 0 then

 Count = Count + 1

 End If

 Input Data

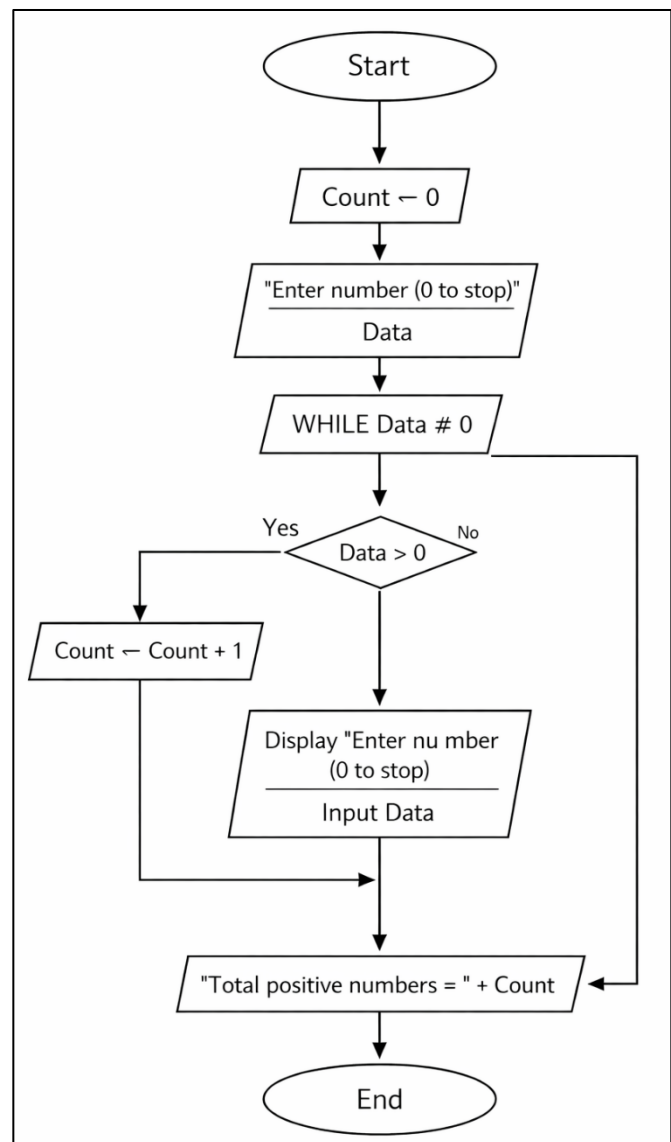
End While

Display Count

Key Idea

- counter variable increases only when the condition is true.

Flowchart:



Activity 4 – Factors & Prime Number

Concept

- Loop + Decision structure

Explanation

- Check every number from 2 to N-1
- If it divides evenly → it is a factor
- If no factor is found → it is a prime number

Pseudocode

Input Number

IsPrime = True

For i = 2 to Number - 1

 If Number % i == 0 then

 Display i

 IsPrime = False

 End If

End For

If IsPrime == True then

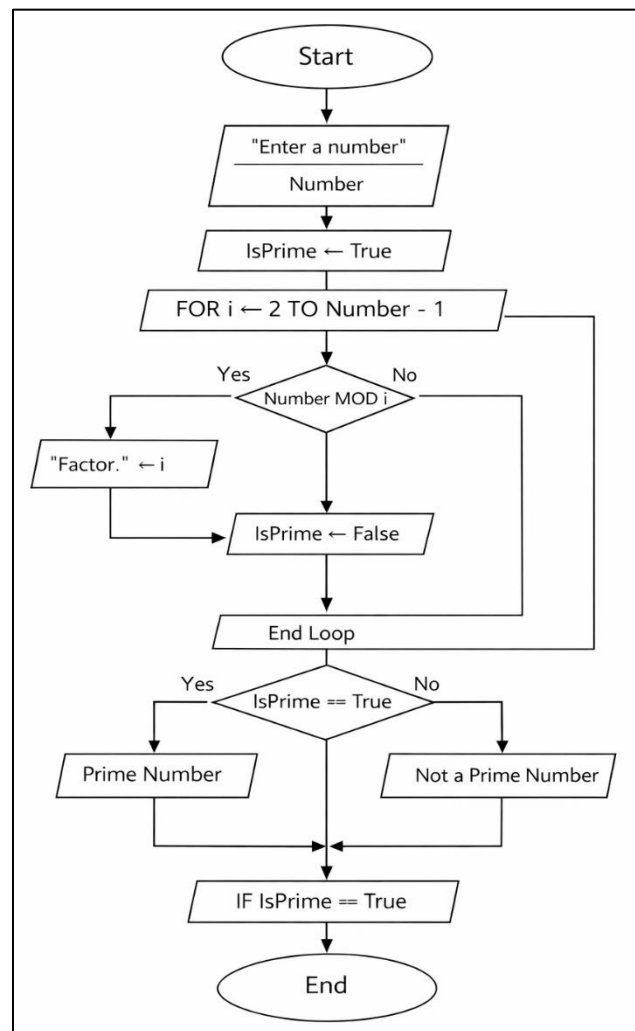
 Display "Prime Number"

End If

Key Idea

- Boolean variable (IsPrime) is used.
- Loop and condition are combined together.

Flowchart:



Graded Lab Task

Lab Task 1 – 5 Day Discount Sale

A Boutique is having a five-day sale. Each day, starting on Monday, the price will drop 10% of the previous day's price. For example, if the original price of a product is \$20.00, the sale price on Monday would be \$18.00 (10% less than the original price). On Tuesday the sale price would be \$16.20 (10% less than Monday). On Wednesday the sale price would be \$14.58; on Thursday the sale price would be \$13.12; and on Friday the sale price would be \$11.81. Develop a solution that will calculate the price of an item for each of the five days, given the original price. Test the solution for an item costing \$10.00.

Problem Understanding

- Each day price decreases by **10% of previous day**
- Loop problem (fixed repetition)

Formula

New Price = Old Price $\times$ 0.9

Algorithm

1. Input original price
2. Set Day = 1
3. Repeat 5 times:
 - Price = Price $\times$ 0.9
 - Display price
 - Increase Day

Pseudocode

Input Price

For Day = 1 to 5

 Price = Price * 0.9

 Display "Day", Day, "Price =", Price

End For

Example (Price = 10)

Day	Calculation	Price
1	10×0.9	9.00
2	9×0.9	8.10
3	8.1×0.9	7.29
4	7.29×0.9	6.56
5	6.56×0.9	5.90

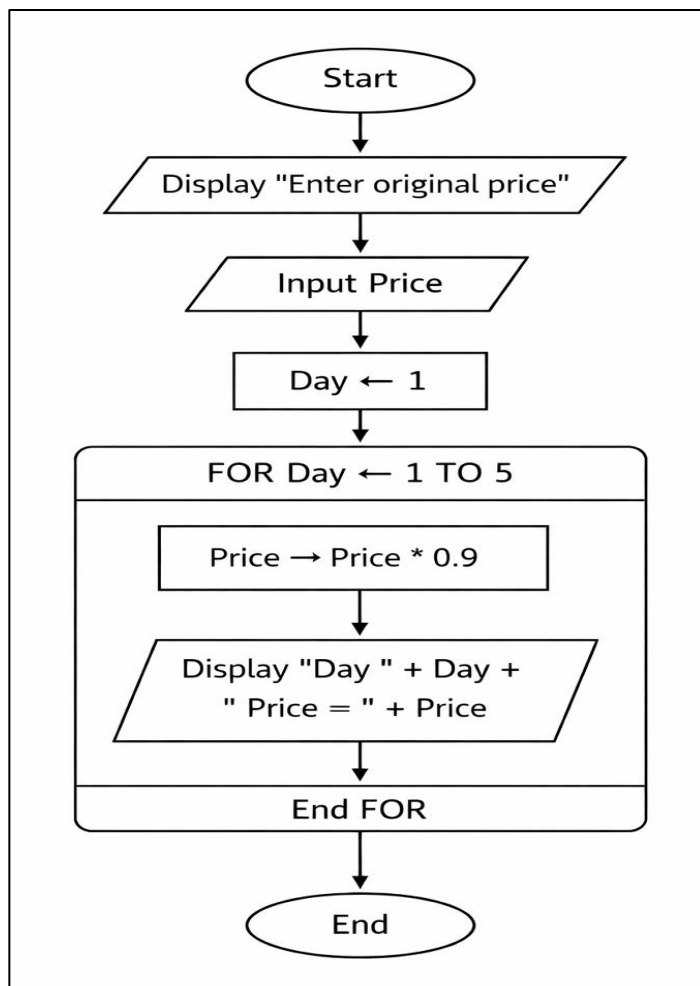
Concept Used

- Counter-controlled loop

Notes

- Initialize price first
- Loop runs exactly 5 times
- Use multiplication (0.9)

Flowchart:



Lab Task 2 – Loan Payment System

Mary Smith, a student, has borrowed \$3,000 to help pay her college expenses. After setting up a budget, \$85 was the maximum monthly payment she could afford to make on the loan. Develop a solution to calculate and print the interest, the principal, and the balance on the loan per month. Other information she would like to know is the number of years and months it will take to pay the loan back and the total interest she will pay during that period. The interest rate is 1% per month on the unpaid balance.

Keep in mind these formulas:

$\text{interest_normal} = \text{balance} * \text{interest_rate}$

$\text{payment} = \text{balance} - \text{interest}$

$\text{new_balance} = \text{balance} - \text{payment}$

Problem Understanding

- Loan = 3000
- Interest = 1% per month
- Monthly Payment = 85

Formulas

$\text{Interest} = \text{Balance} \times 0.01$

$\text{Principal} = \text{Payment} - \text{Interest}$

$\text{New Balance} = \text{Balance} - \text{Principal}$

Algorithm

1. Initialize:

- Balance = 3000
- Payment = 85
- Rate = 0.01
- Months = 0
- Total_Interest = 0

2. Loop while Balance > 0:

- $\text{Interest} = \text{Balance} \times \text{Rate}$
- $\text{Principal} = \text{Payment} - \text{Interest}$
- If $\text{Principal} > \text{Balance}$:
 - a. Adjust last payment
- $\text{Balance} = \text{Balance} - \text{Principal}$
- Add Interest to Total_Interest
- $\text{Months} = \text{Months} + 1$

3. Convert Months into Years + Months

Pseudocode

Balance = 3000

Payment = 85

Rate = 0.01

Months = 0

Total_Interest = 0

While Balance > 0

Interest = Balance * Rate

Principal = Payment - Interest

If Principal > Balance then

Principal = Balance

Payment = Interest + Principal

End If

Balance = Balance - Principal

Total_Interest = Total_Interest + Interest

Months = Months + 1

Display Interest, Principal, Balance

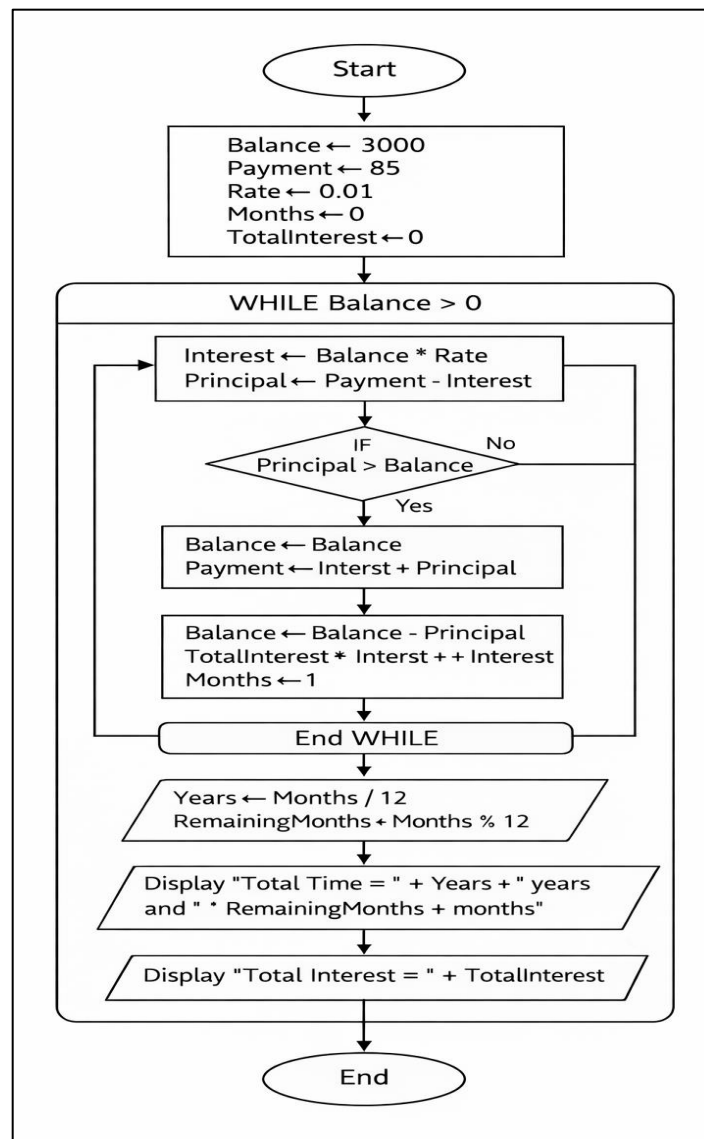
End While

Years = Months / 12

Remaining_Months = Months % 12

Display Years, Remaining_Months, Total_Interest

Flowchart:



Sample Output

Month	Balance	Interest	Principal	New Balance
1	3000	30	55	2945
2	2945	29.45	55.55	2889.45
3	2889	28.89	56.11	2833

Concepts Used

- Sentinel-controlled loop
 - Decision structure
-

Important Notes

I.T.E.M Rule

- Initialize
- Test
- Execute
- Modify

Common Mistakes

- Not updating balance
- Wrong formulas
- Infinite loops
- Ignoring last payment case

Key Learning

- Loops + decisions solve real problems
- Always track:
 1. Balance
 2. Interest
 3. Time

End of Lecture